Course 1 glossary

Terms and definitions from Course 1

A

Advanced provisioning: A cloud delivery model in which the user signs a formal agreement with the cloud service provider and either pays a set price, or is billed monthly

Apache Hadoop: A framework that distributes the processing of data across clusters of computers

Archival policy: An outline of where and how data is stored once an analysis project is complete

Audit: A formal examination of how users are accessing data in order to ensure safe and appropriate access, while identifying and solving any data concerns

Automation: The use of software, scripting, or machine learning to perform data analysis processes without human work

Autoscaling: A cloud service that monitors applications, automatically scaling up or down according to the computing resources needed to meet user demand

B

Backend platform: The components of cloud architecture that make up the cloud, including computing resources, storage, security mechanisms, and management

Business data request: A business question that can be answered with data

C

Check-in: Upload code to a main repository so others can access it and review it

Cloud computing: The practice of using on-demand computing resources as services hosted over the internet

Cloud cost optimization: The process of reducing cloud expenses by implementing cost-reduction strategies

Cloud data analytics: The process of collecting, storing, managing, organizing, transforming, and analyzing large volumes of data using cloud-based services and solutions

Cloud data warehouse: A large-scale, data storage solution hosted on remote servers by a cloud service provider

Collaboration: Members of the data analytics team visit the central system to review data requests, including queries and data delivery methods

D

Data access management: The process of implementing features including password protection, user permissions, and encryption to protect data and related processes

Data analyst: A professional who performs data analysis workflows, including importing, manipulating, calculating, and reporting on business data

Data architect: A professional who collaborates with data analysts, data scientists, and data engineers to design the infrastructure of a database

Data center: A physical building that contains servers, computer systems, and associated components

Data documentation: A written guide to the data contained in a dataset, how it was collected, and how it's organized

Data engineer: A professional who transforms data into a useful format for analysis, and gives it a reliable infrastructure

Data ingestion: The process of obtaining, importing, and processing data for later use or storage

Data integration: The process of combining data from different sources into a single, usable data source

Data lifecycle: The sequence of stages that data experiences, which include plan, capture, manage, analyze, archive, and destroy

Data management: The process of establishing and communicating a clear plan for collecting, storing, and using data

Data pipeline: A series of processes that transport data from different sources to a destination for storage and analysis

Data privacy: Preserving a data subject's information any time a data transaction occurs

Data processing: Raw data is cleaned, organized, and changed into a format for easy analysis

Data retention: The collection, storage, and management of data

Data scientist: A professional who works primarily in data research, identifying business questions, collecting data from multiple sources, organizing it, and finding answers

Data storage: The amount of data kept in storage spaces called "buckets"

Data-collection policy: An outline that creates rules around how data is collected, and what resources may be used in the process

Dataproc: A fully managed service that maximizes open-source data tools for batch processing, querying, streaming, and machine learning

Deletion policy: An outline of when and how data is permanently destroyed

Documentation: A process for preserving all of the details and conversations about each request in one place

Dynamic provisioning: A cloud delivery model in which resources are adjusted based on the user's changing needs, and they are only charged for what they use

E

Encryption: The process of encoding information

F

Frontend platform: The component of cloud architecture that users interact with

G

Google BigQuery: A data warehouse on Google Cloud used to query data, filter large datasets, aggregate results, and perform complex operations

H

Historical records: Past data or information that's been stored and can be accessed or retrieved when needed

Hold: A policy placed on a dataset that prevents its deletion, or prevents deletion capabilities for certain accounts

Hybrid cloud: A cloud model that combines public and private models, so organizations can enjoy both cloud services, and the control features of on-premises cloud models

I

Identity Access Management (IAM): The process of assigning certain individual roles or groups with access to particular resources

Infrastructure as a service (IaaS): A cloud computing model that offers on-demand access to IT infrastructure services, including hardware, storage, networking, and virtualization tools

M

Managed service: The act of a third-party provider taking care of the ongoing maintenance, management, and support of an organization's cloud infrastructure and applications

Measured service: A service that charges users only for what they use, based on the number of transactions, the storage volume, and the amount of data transferred

N

Network use: The amount of data that is read or moved between storage buckets

O

On-premises: Information technology infrastructure that is physically located in an organization's own data center or office

P

Parent-child relationship: A feature of a ticketing system that enables users to divide a single ticket into subtickets, which can be worked on simultaneously

Personally identifiable information (PII): Data that is unique to an individual, and therefore can be used to distinguish one person from another

Platform as a service (PaaS): A cloud computing model that provides hardware and software tools to create an environment for the development of cloud applications, simplifying the application-development process

Private cloud: A cloud model that dedicates all cloud resources to a single user or organization, and is created, managed, and owned within on-premises data centers

Protected health information (PHI): Health data that can identify an individual, like information about patient demographics, mental or physical health diagnoses or treatments, and payment records related to health care

Public cloud: A cloud model that delivers computing, storage, and network resources through the internet

R

Refactoring: A cloud-migration strategy that involves building all-new applications from scratch and discarding old applications

Rehosting or "lift and shift": A cloud-migration strategy that involves moving an entire on-premises system to the cloud

Replatforming: A cloud-migration strategy that involves making small changes to the on-premises system once it's migrated to the cloud

Repurchasing: A cloud-migration strategy that involves moving applications to a new, cloud-based service platform, usually a software-as-service

Reserved instances: A cloud payment model in which an organization purchases a specific amount of resources for a certain time period, and receives a discount in return for this commitment

Resource provisioning: The process of a user selecting appropriate software and hardware resources, and the cloud service provider setting them up and managing them while in use

Retention policy: An outline of how data is archived and deleted

Retiring: Applications that are no longer useful are turned off

Rightsizing: The process of adjusting computing resources to fit the exact needs of an application or workload

S

Security key: An authentication method that uses a physical digital signature, or key, to verify a user's identity before accessing specific resources

Self-provisioning: A cloud delivery model in which the user purchases resources from the cloud provider through a website or online portal, then the resources are quickly made available for the user

Software as a service (SaaS): A cloud computing model that provides users with a licensed subscription to a complete software package

T

Traditional computing: A computing model that enables data storage, access, and management through the use of physical hardware and software within a network infrastructure, typically located on-premises

U

Uptime: The amount of time a machine is operational

V

Versioning: The process of creating a unique way of referring to data

Virtualization: Technology that creates a virtual version of physical infrastructure, like servers, storage, and networks